Kush Rishi

GNSS Analyst at Xona. BSc in Geomatics Engineering, University of Calgary.

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Experience

GNSS Analyst · Xona

June 2026–present

Develop Python and Linux workflows for GNSS collection, processing, monitoring and validation. Build data-quality checks, engineering metrics and dashboards; support receiver integration and troubleshooting.

Research Assistant · University of Calgary

December 2025–April 2026

Processed LiDAR and photogrammetric point clouds in CloudCompare, calculated spatial quality metrics and prepared documented datasets and visualizations for faculty research.

Geomatics Team Member · Relectric Car Team

September 2023–April 2026

Supported electric-vehicle conversion with spatial measurements and geometry models. Worked with GNSS, IMU and LiDAR data and prepared technical outputs for other subteams.

Survey Assistant · McElhanney

May–August 2024

Supported control, layout and as-built surveying for the Valley Line West LRT extension using GNSS receivers, total stations and digital levels.

Software Application Developer · Hycroft Chiropractic & Massage

September 2021–June 2022 · Freelance

Built and maintained the Stretch2Go application in Dart, including booking, exercise plans, journaling and content updates. Refined user flows through testing and feedback.

Education

University of Calgary

BSc Geomatics Engineering, With Distinction · 2022–2026

President, Geomatics Engineering Students' Society; Engineering Students' Society Board of Directors, 2025–2026.

Selected projects

Capsule Review

Built a local capsule-endoscopy viewer and fixed-budget frame-selection experiments. In a locked study of 163,161 images from ten examinations, mean blood-positive annotation-run recall at 10% workload was 49.35% for score-per-bin selection versus 50.62% for uniform sampling across six positive-run examinations. Annotation runs are not independently adjudicated lesions; the public demo uses synthetic images and scores.

TrueMargin

Evaluated registration variability against known spatial error in 30 controlled cases across ten anatomies, followed by separate calibration and evaluation cohorts. Median anatomy-level Spearman correlation was 0.6841 for spread versus 0.7203 for inverse-consistency error; six case-level associations were negative. Some calibrated bounds were very large. External landmark testing remains unfinished.

Model Regression Forensics

Built a prediction importer, release-policy checks and local reports. In two constructed Banking77 benchmarks, a simple label-overlap baseline ranked the injected change first; learned diagnostics did not consistently match it. A separate example shows how different repairs can fit the same observed regression.

Autonomy Simulation Lab

Built a browser planning/localization simulator and a separate C++/Python recording-replay tool with CPU inference and configuration comparison. A crop-aware optimization reduced median whole-process time by 8.97% across three matched 108-frame pairs with exact recorded output parity on one Linux host. Measurements apply to that host, model and workload.

Interactive demo

Technical focus

Python, C++, PyTorch, Linux, Git, CI, data pipelines, numerical evaluation, GNSS/PNT, localization, state estimation and image registration.